15 April 2026 09:56

DevOps:

Development team :- create an application -
:-hosting
Serving the application through the web

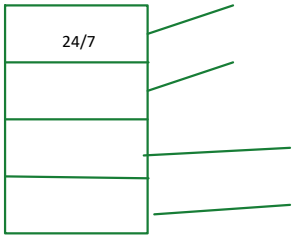
Virtual machine

By using service providers.. AWA, Azure,GCP
Once your requirement is completed -> you terminate servers
Pay As you Go

CPU,
Memory
Disk space

Server1-- login, run command get details

Server2 -- login , run commands get the details .
....10th server 1hr 1.30



www.flipkart.com

Development team

DevOps Team

10month

30lak 10%

Introduce python ...

-> a real devops en does not depend on repetitive works manually
-> A real Devops eng AUTOMATES
-> and python is the tool that help us do this automation

--> python is not just a programing language and it is a tool to control servers, automate the tasks , and manage infrastructure

Python .exe download , 3.11.

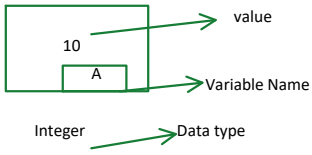
Vscode , IDE

Reason ----

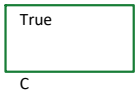
Simple syntax	Reads almost like English language
Huge Library Ecosystem	Libraries for AWS, Docker,Kubernates, Ansible, everything
Cross-platform	Runs on Linux, masOS,windows unchanged
Community	Millions of Developers , thousands of Devops tutorials
Automation Power	Automate files Ops ,API call, cloud tasks with few lines

Variables :

A=10

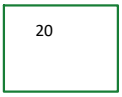


c= True



C
Bool

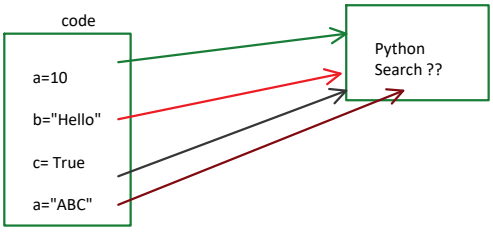
d=20



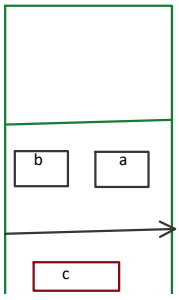
int

b='hello'

Str



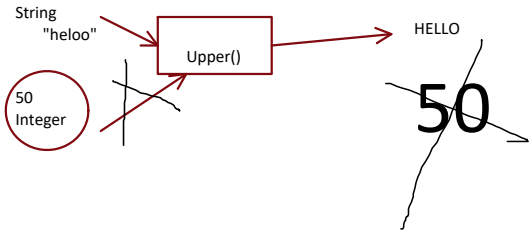
Memory Location



If you update the value for the variables then , old space is going to delete and the new space is created based on the data type

Python automatically detect the Data types
Dynamic :the data types can change any time

Why data types?



Prevent problems

Int	int

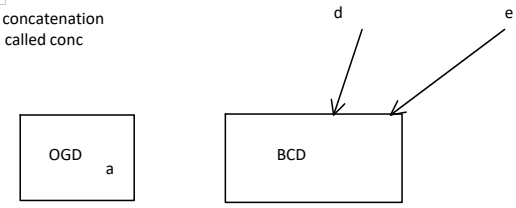
2 + 3 --> 5
Addition

Str	Str

"2" + "3" --> "23" concatenation
Joining the text is called conc

a="BCD"
a="OGD"

d="BCD"
e="BCD"



Data Types

No value

None
Nothing

Single value (primitive data type)
It contains only one value

Int	Flot	Str	Bool
50	3.250	"Devops"	True

Multiple value
Collections, data structures, containers

List	Set	Tuple	dict
[1,2,3]	{1,2}	a=('a','b','c')	{'a':10,'b':20,'c':30 }

Mutable:- objects can be modified 'in-place' this means you can add, remove,update their content without changing object unique identification or memory address
Immutable:- objects cannot be changed once it is created .

Data types	Description
Int(integer)	**Whole numbers : eg- port=8080**
Float(Decimal)	**Numbers with decimal point eg- CPU_usage =73.5**
Str(String)	**Text enclosed with Quotes eg- server='nginx'**
Bool(Boolean)	**True or False only eg-: is_running=True**
List	**Ordered collection eg- ["web1","db1","nginx"]**
Dict(dictionary)	**Key-value pairs . Eg-{'ip':'10.0.0.1','port':80}**
Tuple	**Immutable ordered collection eg-('ubuntu','22.04')**
None	**It represents no value or empty , eg-result= none**

Operators :

Operator type	Symbols and meaning
Arithmetic	+ add, - subtraction , * multiplication , / divide, // floor div, % modulus , ** power
Comparison	== equal , != not equal , > greater than , < less, >= greater - or equal, <== less-or-equal
Logical	and (both true) , or (either true), not(opposite)
Assignment	= assign , += add assign , -= subtract and assign

Values are objects of data type class -->

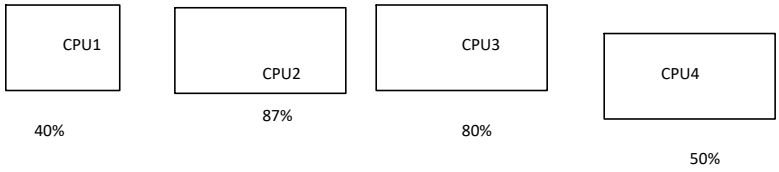
Control flow

Control flow means deciding WHAT python should do based in conditions … it checks condition , if it is true , run this block, elif(elseif) check another condition if the first condition was false .

If(condition 1) == true
{
Execute this block
}when condition 1 false then elif(execute)
Else run when none of the above condition were true

#type1
If (condition1) = true
{
Execute this block
}
Else {(when condition1 =False)
 execute else block
}
Type 2:
If(condition 1=True)
{
Execute this block
} elif(condition)=true
{
Execute elif block
}
Else {
execute none of the above false
}

Subject	Marks	Status(If marks>=35) pass else fail
Subject1	40	Pass
Subject2	30	Fail
Subject3	55	Pass
Subject4	35	Pass
Subject5	20	Fail



AUTO SCALING :
INCREASING OR DECREASING SERVERS BASED ON LOAD AUTOMATICALLY

Situation 1:

If 87.3>=90 #-false

Elif 87.3>=75 True

```
cpu_usage=30 # simulated current
```

```python
cpu_usage=30 # simulated current CPU usage
if cpu_usage>=90:
    print("CRITICAL:CPU usage is dangerously high")
    print("ACTION: triggering auto scale ")
elif cpu_usage>=75:
    print("WARNING: CPU usage is elevated ")
    print("ACTION:  Sending alert to slack ...")
elif cpu_usage>=50:
    print("INFO : CPU usage is moderate.")
else:
    print("OK: CPU usage is normal ")
```

Situation 1:

If 87.3>=90 #-false

Elif 87.3>=75 True

```
WARNING: CPU usage is elevated
ACTION:  Sending alert to slack …
```

Situation 2 :

If 60>=90 false
elif 60>=75 false

Elif 60>=50 True

```
INFO : CPU usage is moderate.
```

Situation 3:

If 30>=90 false
Elif 30>=75 false
Elif 30>=50 false
Else ---
OK: CPU usage is normal